Autonomous Smart Parking & Revenue Management System

SWE 4740 – Embedded Software Development Lab



Team-05



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Problem Statement



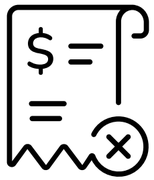
No Real-Time Monitoring



No Secure Access Control



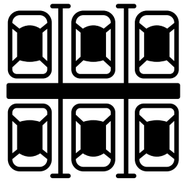
Manual Time Tracking



Error-Prone Billing

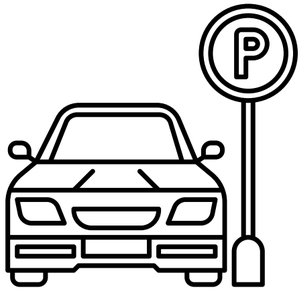


Traffic Congestion



Poor Space Utilization

System Features



Parking Slot Monitoring

- Ultrasonic sensors per slot
- Green LED = Empty | Red LED = Occupied
- Real-time updates to LCD



Automatic Entry Gate

- Servo motor as physical barrier
- Opens only: slot available + auth passed
- PWM-controlled smooth operation

System Features



Parking Time Calculation

- Timer starts on vehicle entry
- Stops on exit automatically
- Stored for accurate billing



Billing System

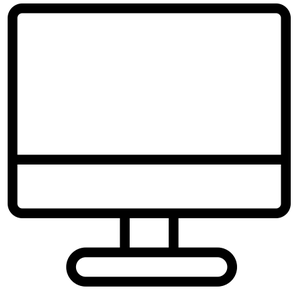
- First 1 hour = 50 Taka
- Each additional hour = 30 Taka
- Auto-calculated, shown on LCD

System Features



Security System

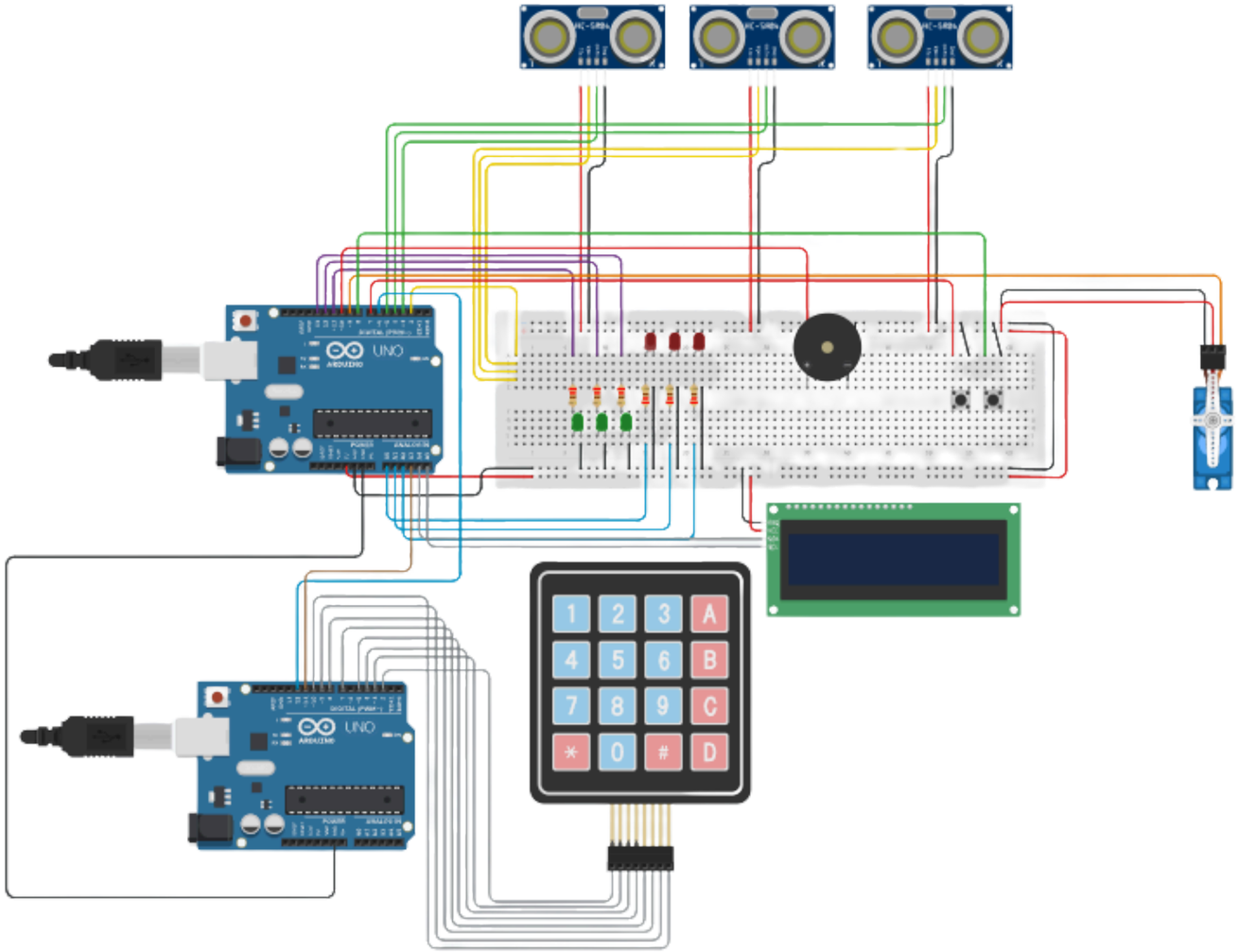
- Keypad password authentication
- Buzzer alert after 3 failed attempts
- Failed attempts logged



LCD Display + Over-Capacity

- Shows slots, fee, system messages
- Displaying 'PARKING FULL' when full
- Keypad locked at full capacity

System Design



Component Selection

Arduino Uno R3

Central microcontroller — coordinates all sensors, actuators & display

4×4 Keypad

User authentication — password entry for gate access

Push Buttons

Simulate vehicle entry and exit events in the system

Ultrasonic Sensors

Detect presence/absence of vehicles in each parking slot

Green & Red LEDs

Visual slot indicators — green = free, red = occupied

16×2 LCD Module

Displays slot count, availability, assigned slot & billing fee

Servo Motor

Physical entry gate — PWM-controlled open/close mechanism

Buzzer

Security alert — triggered after 3 consecutive failed auth attempts

System Workflow

1 Vehicle Arrives

Driver approaches; system checks slot availability via ultrasonic sensors.

2 Authentication

User enters keypad password. 3 failures → buzzer alert. If full → gate locked.

3 Gate Opens

On success + available slot: servo opens gate. Timer starts automatically.

4 Slot Assigned

LCD shows assigned slot. Red LED activates. Slot count updates in real time.

5 Vehicle Exits

Exit button pressed. Timer stops. Total duration calculated instantly.

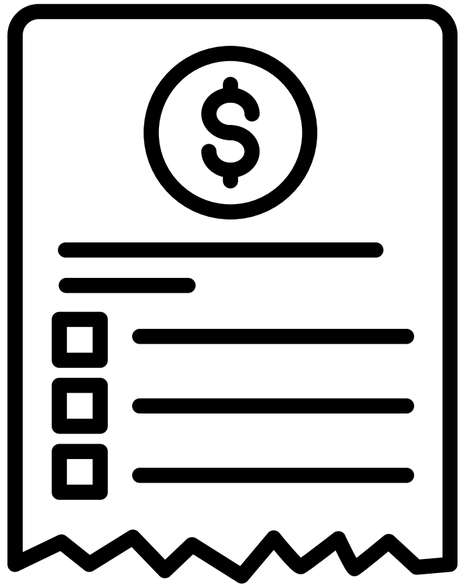
6 Fee Displayed

Billing: 50 Taka/1st hr + 30 Taka/additional hr. Fee shown on LCD.

Billing System

- First 1 Hour = 50 tk
- Additional Hours = 30 tk/hour

Total Bill = 50 + (N-1) * 30



Security System



Keypad Authentication

Password must be entered before gate opens.



Attempt Logging

Failed attempts recorded for security audit trail.



3-Strike Lockout

Buzzer triggered after 3 consecutive wrong attempts.



Over-Capacity Lock

Keypad disabled & scrolling 'FULL' message when all slots occupied.

Challenges Faced

Sensor Calibration

Ultrasonic sensors needed careful tuning to avoid false positives for varying vehicle sizes.

Gate Control Reliability

Ensuring smooth PWM servo operation under all conditions without mechanical delays.

Real-Time LCD Updates

Efficiently managing data flow to keep LCD refreshed without blocking core logic.

Security Edge Cases

Handling repeated failures, lockout logic, and preventing accidental lockouts.

Timing & Synchronization

Entry/exit events had to be precisely timed to prevent incorrect duration and billing errors.

Multi-Vehicle Handling

System had to track multiple slots independently without conflicts in assignment or fees.

Component Integration

Coordinating sensors, LEDs, servo, buzzer & LCD without signal interference or timing conflicts.

Simulation Limitations

Some hardware behaviors in Tinkercad/Proteus differ from real-world — required adjustments.

Future Improvements



Dynamic Pricing

Algorithmic rates during peak hours (>80% occupancy) to optimize revenue.



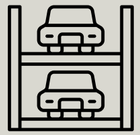
Diagnostic Serial Monitor

'Technician Mode' for daily revenue, sensor health & error logs via serial.



Power-Saving Mode

Interrupt-driven sleep mode during idle hours to reduce energy consumption.



Multi-Level Parking

Expand to multiple floors with per-level slot allocation and inter-MCU comms.



Mobile App Integration

Remote slot checking, reservation, and digital payment via mobile app.

Thank You